GYM AI FITNESS PLANNER

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COURSE – B.TECH

SLOT – C11+C12+C13

1. Introduction

In today's fast-paced world, maintaining physical health and a proper diet is incredibly important. However, many beginners struggle to figure out the right diet and workout plan for their specific body types. This project focuses on building an AI-powered fitness planning system that generates customized fitness and diet plans using Python and machine learning models. The project includes a login and signup system with local data storage so that users can securely access and update their fitness profiles, making it a complete, real-world applicable solution.

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2. Objective

The main objectives of this project are:

- To build a Python application that generates personalized fitness and diet plans using machine learning.
- To accurately calculate body metrics like BMI and predict daily calorie requirements using a Linear Regression model.
- To classify optimal workout plans using a Decision Tree Classifier.
- To create a functional command-line based interface.
- To implement a basic login and signup system using JSON for data persistence.
- To store, display, and manage user prediction history across multiple sessions.

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3. Problem Statement

Creating a structured fitness journey manually can be confusing and overwhelming. Different factors like age, weight, height, BMI, and dietary preferences require specific adjustments to calorie intake and exercise routines. This project aims to solve this problem by using machine learning models trained on health data to automatically process these input values and provide users with a clear, accurate, and actionable health plan.

4. Methodology

The project is implemented step by step as follows:

- First, the application trains two ML models at startup using predefined health data: a Linear Regression model for calorie prediction and a Decision Tree Classifier for workout plan classification.
- The application initializes by loading existing user data from a local JSON file (users.json).
- A menu-driven interface is presented, allowing the user to sign up, log in, or exit the system.
- Once authenticated, the user enters their body metrics (age, weight, height) and selects their diet type (Vegetarian/Non-Vegetarian) and specific fitness goal (Weight Loss, Muscle Gain, Maintain).
- The system calculates BMI and uses the trained ML models to predict calories and classify the best workout plan.
- The system selects the appropriate diet plan from a pre-defined goal and diet-type mapping.
- The generated plan is displayed to the user and saved back to the JSON file for future access.
- The dashboard also gives the user the ability to view their previous data or delete their saved records entirely.

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5. Tools and Technologies Used

- Python 3
- NumPy (for array handling and feature vectors)
- scikit-learn (for LinearRegression and DecisionTreeClassifier)
- JSON module (for data storage)
- OS module (for file handling)

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6. Features of the System

- User login and signup functionality.
- BMI calculation based on height and weight.
- AI-predicted daily calorie requirement using Linear Regression.
- AI-predicted workout plan using a Decision Tree Classifier.
- Personalized diet plans categorized into Vegetarian and Non-Vegetarian.
- Customized workout routines tailored for Weight Loss, Muscle Gain, or Maintenance.
- Persistent storage of user data using JSON.
- Option to view previously generated plans or clear data history

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7. System Design

The system works in the following flow:

- ML models (LinearRegression + DecisionTreeClassifier) are trained at startup on built-in health data.
- User signs up or logs in via the main menu.
- Saved user data is loaded from the JSON file.
- User selects an option from the dashboard (View Data, Generate Plan, Delete Data).
- System prompts for physical metrics and personal goals.
- System computes BMI, feeds it as a feature into the ML models, and retrieves predictions.
- The finalized data is saved back to the system and displayed on the screen.

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8. Logic Used

Instead of a machine learning algorithm, this project utilizes

Unlike simple rule-based or formula-driven systems, this project uses trained machine learning models for its core predictions.

8.1 Training Data

A hardcoded dataset of 15 samples is used for training, where each sample contains:

- Features: Age, Weight (kg), Height (cm), BMI, Goal Encoded (0=Weight Loss, 1=Muscle Gain, 2=Maintain)
- Calorie Labels: Target daily calorie values (CALORIE_Y)
- Workout Labels: Workout plan category index 0–4 (WORKOUT_Y)

8.2 Calorie Prediction – Linear Regression

A LinearRegression model (from scikit-learn) is trained to predict the daily calorie intake based on the user's feature vector. The model learns the relationship between physical metrics and calorie needs.

8.3 Workout Classification – Decision Tree

A DecisionTreeClassifier (max_depth=4) is trained to map user features to one of 5 workout plan categories:

- 0 – Light Cardio (3 days/week): Walk/jog, yoga, stretching
- 1 – Cardio + Strength (4 days/week): 20 min cardio + full body weights
- 2 – Strength Focus (5 days/week): Upper/Lower split, progressive overload
- 3 – Push Pull Legs (6 days/week): Chest/Shoulders, Back/Biceps, Legs/Core
- 4 – Balanced Routine (4 days/week): 2 strength + 2 cardio days

8.4 Diet Plan Matching

Diet plans are matched from a dictionary keyed on (goal_encoded, diet_type). Both Vegetarian and Non-Vegetarian variants are available for Weight Loss (0), Muscle Gain (1), and Maintain (2) goals.

8.5 BMI Calculation

BMI = Weight (kg) / (Height in meters)^2

BMI Categories: Underweight (<18.5), Normal (18.5–24.9), Overweight (25–29.9), Obese (≥30)

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9. Input and Output

Input:

- Age
- Weight (in kg)
- Height (in cm)
- Diet Preference (Veg / Non-Veg)
- Fitness Goal (Weight Loss / Muscle Gain / Maintain)

Output:

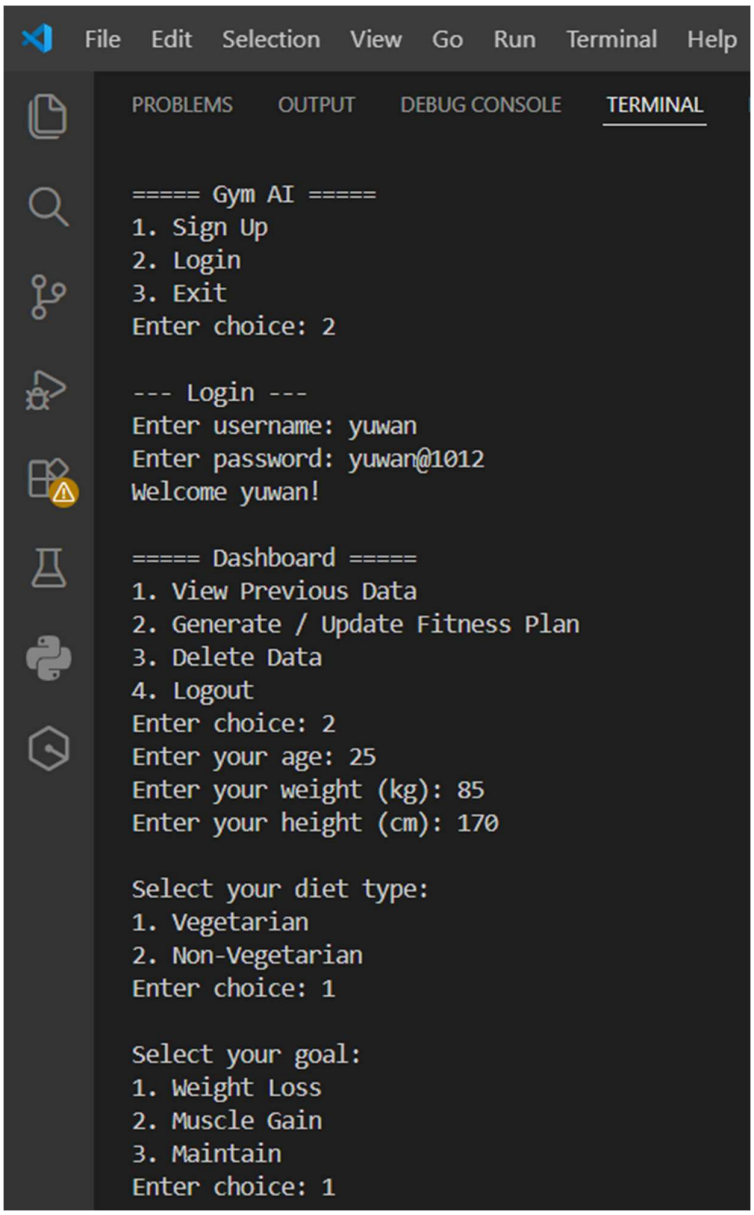
- BMI (Body Mass Index) and BMI Category
- AI-Predicted Daily Calorie Requirement (via Linear Regression)
- AI-Assigned Diet Plan (Vegetarian or Non-Vegetarian)
- AI-Assigned Workout Plan (via Decision Tree Classifier)

10. Sample Result

Example:

Input:

- Age: 25
- Weight: 85 kg
- Height: 170 cm
- Diet Type: 1 (Vegetarian)
- Goal: 1 (Weight Loss)



Output:

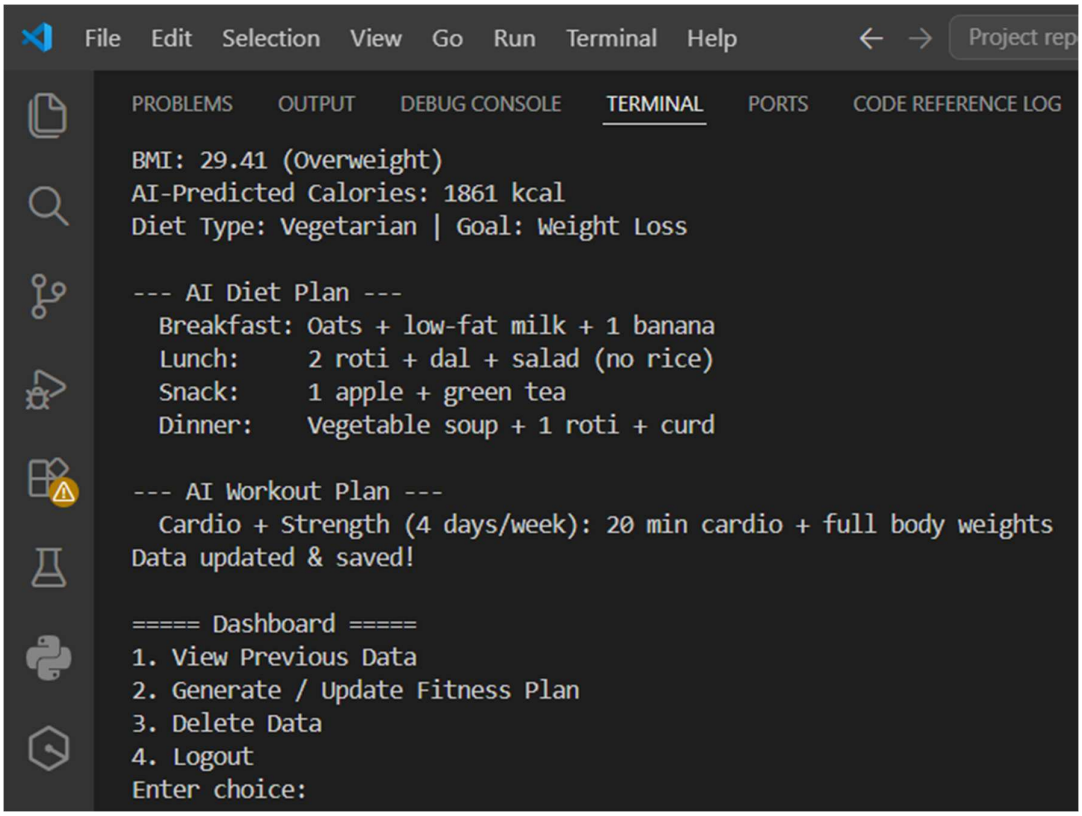
- BMI: 29.41 (Overweight)
- AI-Predicted Calories: ~1800 kcal
- Diet Type: Vegetarian | Goal: Weight Loss

Diet Plan:

- Breakfast: Oats + low-fat milk + 1 banana
- Lunch: 2 roti + dal + salad (no rice)
- Snack: 1 apple + green tea
- Dinner: Vegetable soup + 1 roti + curd

Workout Plan:

- Cardio + Strength (4 days/week): 20 min cardio + full body weights



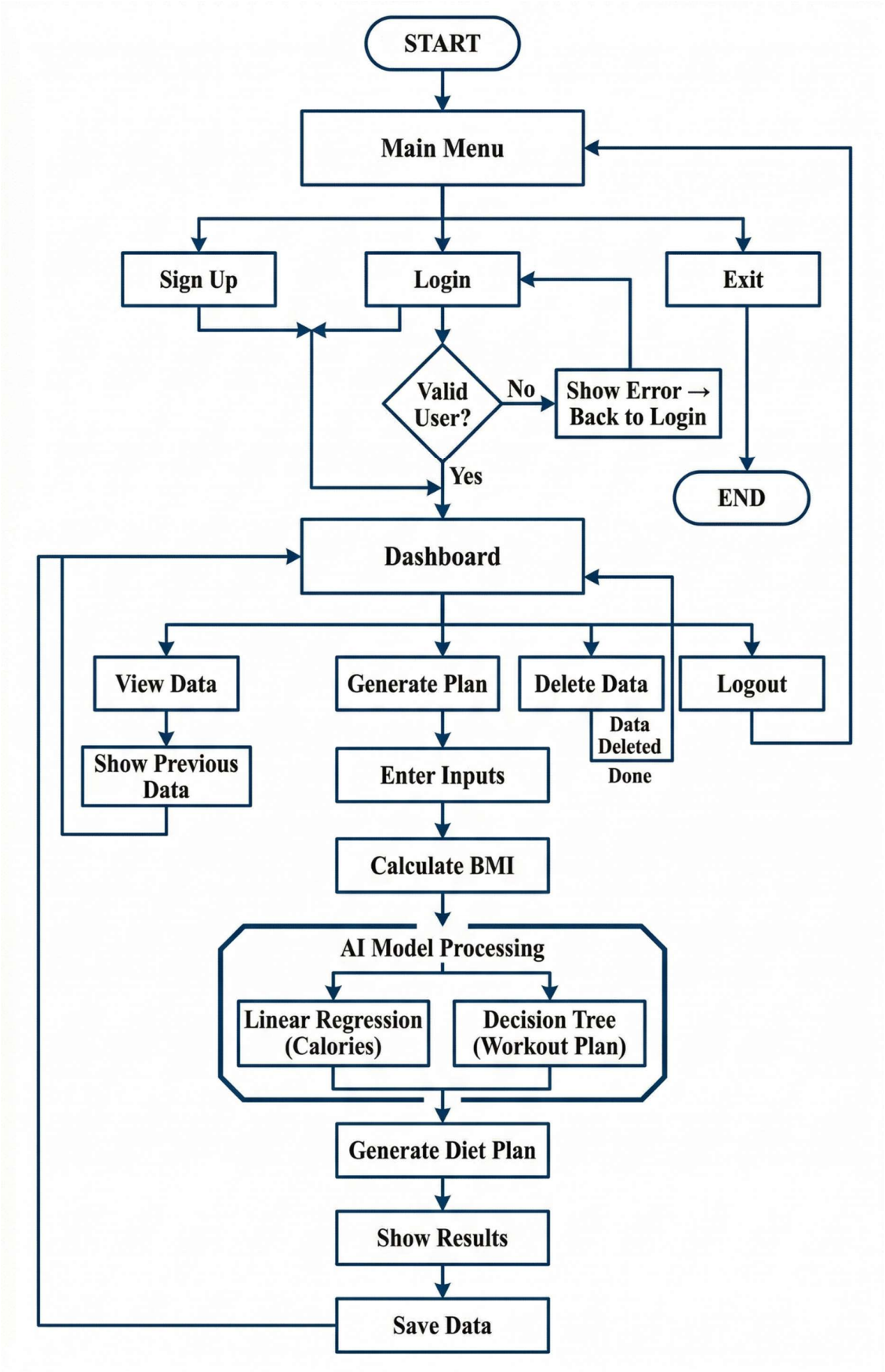
11. Advantages

- Uses real ML models (Linear Regression + Decision Tree) for smarter, data-driven predictions.
- Easy to use, beginner-friendly interface.
- Fast and instantaneous metric calculations and predictions.
- Includes a robust login system.
- Stores user data persistently so it is not lost after closing the app.
- Supports both Vegetarian and Non-Vegetarian diet plans for all three fitness goals.

12. Limitations

- Training dataset is small (15 samples), which may limit prediction accuracy.
- No graphical user interface (GUI).
- Passwords are stored in a simple, unencrypted text format inside the JSON file.
- The ML models cannot dynamically learn from new users' data over time (no online learning).
- No cross-validation or test-train split is applied to evaluate model performance.

DIAGRAM OF THE MODEL



13. Future Scope

- Expand the training dataset with real-world health records to improve ML model accuracy.
- Implement model persistence using joblib/pickle so models are not retrained on every run.
- Create a graphical or web interface using Streamlit, Flask, or a mobile app framework.
- Improve security using proper password hashing/encryption (e.g., bcrypt).
- Add online learning to allow the model to adapt to new user data over time.
- Include daily water intake tracker and macro-nutrient (Carbs, Protein, Fat) breakdown.

14. Conclusion

This project demonstrates how Python and machine learning can be applied to solve real-life problems like personal health tracking and fitness planning. By integrating scikit-learn's Linear Regression and Decision Tree Classifier, the system moves beyond simple rule-based logic and delivers data-driven, personalized fitness recommendations. The addition of login features and persistent history makes it an interactive and genuinely useful utility that reflects the practical application of AI in health and wellness.

15. References

- Python Official Documentation – https://docs.python.org
- scikit-learn Documentation – https://scikit-learn.org
- NumPy Documentation – https://numpy.org/doc
- JSON Module Documentation – https://docs.python.org/3/library/json.html